



2024

HIGHER SCHOOL CERTIFICATE TASK 2

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
45

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section

Section II – 40 marks (pages 4 – 7)

- Attempt Questions 6 – 8
- Allow about 1 hour and 22 minutes for this section

Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 5.

1 Which of the following is an expression for $\int \cos^2 4x \, dx$?

- A. $\frac{x}{2} + \frac{1}{8} \sin 8x + C$
- B. $\frac{x}{2} - \frac{1}{8} \sin 8x + C$
- C. $\frac{x}{2} + \frac{1}{16} \sin 8x + C$
- D. $\frac{x}{2} - \frac{1}{16} \sin 8x + C$

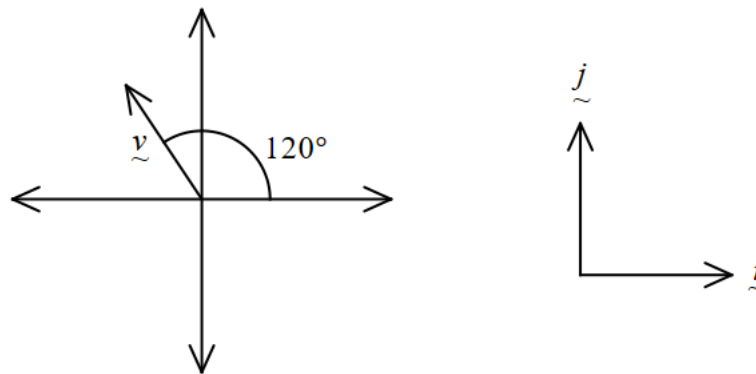
2 Which of the following is equivalent to $3 \sin x - 4 \cos x$, to the nearest degree?

- A. $5 \cos(x + 143^\circ)$
- B. $5 \cos(x - 143^\circ)$
- C. $5 \sin(x + 143^\circ)$
- D. $5 \sin(x - 143^\circ)$

3 What is the derivative of $\cos^{-1}(9x^2)$?

- A. $\frac{-18x}{\sqrt{1-81x^4}}$
- B. $\frac{-18x}{\sqrt{1-81x^2}}$
- C. $\frac{-6x}{\sqrt{1-9x^4}}$
- D. $\frac{-6x}{\sqrt{1-9x^2}}$

- 4 Consider the vector $\underline{v}$ shown below. It is known that $|\underline{v}| = 2$.



Which of the following vectors is perpendicular to the vector $\underline{v}$?

- A. $\underline{a} = -\underline{i} - \sqrt{3}\underline{j}$
 - B. $\underline{b} = -\sqrt{3}\underline{i} + \underline{j}$
 - C. $\underline{c} = \underline{i} + \sqrt{3}\underline{j}$
 - D. $\underline{d} = \sqrt{3}\underline{i} + \underline{j}$
- 5 Given two non-zero vectors $\underline{a}$ and $\underline{b}$, $|\underline{a} + \underline{b}| = |\underline{a} - \underline{b}|$. Which of the following must be true?
- A. $|\underline{a}| = |\underline{b}|$
 - B. $\underline{a}$ is perpendicular to $\underline{b}$
 - C. $\underline{a}$ is parallel to $\underline{b}$
 - D. $|\underline{a}| + |\underline{b}| = |\underline{a}| - |\underline{b}|$

Section II

40 marks

Attempt Questions 6 – 8

Allow about 1 hour and 22 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 6 – 8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks) Use the Question 6 Writing Booklet.

(a) Evaluate $\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{16-9x^2}}$. **3**

(b) Solve $7\sin\theta - 2\cos\theta = 5$, for $0^\circ \leq \theta \leq 360^\circ$, by using the substitution $t = \tan\frac{\theta}{2}$. **3**
Write the solutions correct to the nearest minute.

(c) Consider the vectors $\underline{a} = 5\underline{i} - 3\underline{j}$ and $\underline{b} = 3\underline{i} + 4\underline{j}$.

(i) Find $\underline{a} \cdot \underline{b}$. **1**

(ii) Find the vector projection of $\underline{b}$ onto $\underline{a}$. **2**

(d) (i) Express $\sqrt{3}\cos x - \sin x$ in the form $A\cos(x + \alpha)$, where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. **2**

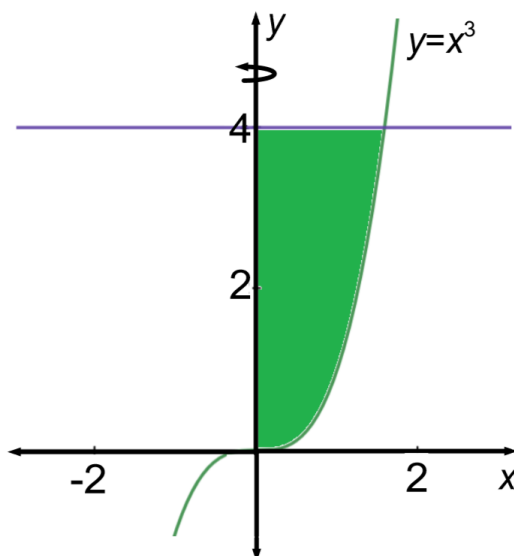
(ii) Hence, or otherwise, solve the equation $\sqrt{3}\cos x - \sin x = -1$ for $-\pi \leq x \leq \pi$. **2**

End of Question 6

Question 7 (13 marks) Use the Question 7 Writing Booklet.

- (a) In the diagram below, the shaded region bounded by the curve $y = x^3$, y -axis and the line $y = 4$, is rotated about the y -axis.

3

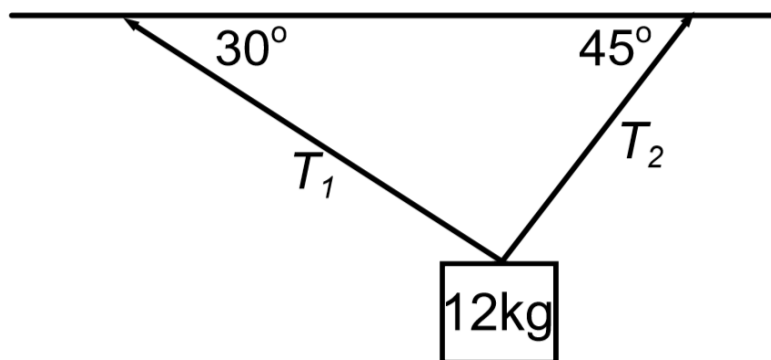


Find the volume of the solid of revolution correct to 1 decimal place.

- (b) Solve $2 \cos^2 2x = 6 \sin^2 x - 1$, for $0 \leq x \leq 2\pi$.

3

- (c) A particle of mass 12kg is suspended by two strings attached to two points in the same horizontal plane. The two strings make angles of 30° and 45° respectively to the horizontal. Let the tensions on the string be T_1 and T_2 newtons as shown.



- (i) Show that $T_1 = \frac{\sqrt{6}}{3} T_2$.
- (ii) Hence, or otherwise, find the value of T_1 to the nearest newton.
Let the acceleration due to gravity be 10 m/s^2 .

1

2

Question 7 continues on page 6

(d) (i) Show that $\frac{d}{dx} x \tan^{-1}\left(\frac{x}{2}\right) = \frac{2x}{4+x^2} + \tan^{-1}\left(\frac{x}{2}\right)$. **2**

(ii) Hence, find $\int \tan^{-1}\left(\frac{x}{2}\right) dx$. **2**

End of Question 7

Question 8 (14 marks) Use the Question 8 Writing Booklet.

- (a) Find $\int \cos 4x \cos x \, dx$. **2**
- (b) Use the substitution $u = x^2 + 5$ to evaluate $\int_0^2 \frac{2x^3}{\sqrt{x^2 + 5}} \, dx$, leaving your answer as an exact value in simplest form. **3**
- (c) Use mathematical induction to prove that $2^n + 3^n - 5^n$ is divisible by 6 for all positive integers n . **3**
- (d) (i) If $t = \tan x$, show that $\tan 3x = \frac{3t - t^3}{1 - 3t^2}$. **2**
- (ii) If $\tan 3x \tan x = 1$, show that $t^4 - 6t^2 + 1 = 0$. **1**
- (iii) By considering $x^4 - 6x^2 + 1 = 0$ and using the result of (ii), deduce that $\tan \frac{\pi}{8} = \sqrt{3 - 2\sqrt{2}}$. **3**

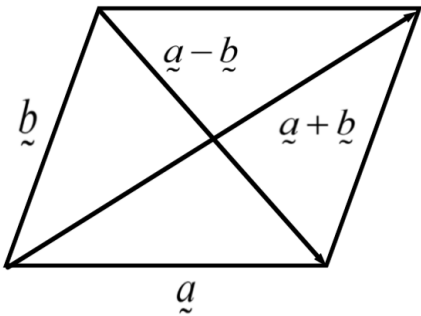
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BAULKHAM HILLS HIGH SCHOOL

2024 YEAR 12 EXTENSION 1 TASK 2 SOLUTIONS

Suggested Solutions		Marks
Section I		
1. C	$\int \cos^2 4x \, dx = \frac{1}{2} \int (1 + \cos 8x) \, dx$ $= \frac{1}{2} \left(x + \frac{1}{8} \sin 8x \right) + C$ $= \frac{x}{2} + \frac{1}{16} \sin 8x + C$	1
2. B	$5 \cos(x - 143^\circ) = 5 \cos x \cos 143^\circ + 5 \sin x \sin 143^\circ$ $= -4 \cos x + 3 \sin x$ $= 3 \sin x - 4 \cos x$	1
3. A	$\frac{d}{dx} \cos^{-1}(9x^2) = -\frac{18x}{\sqrt{1 - (9x^2)^2}}$ $= -\frac{18x}{\sqrt{1 - 81x^4}}$	1
4.D	$\underline{y} = 2 \cos 120^\circ \underline{i} + 2 \sin 120^\circ \underline{j}$ $= -\underline{i} + \sqrt{3} \underline{j}$ The vector $a\underline{i} + b\underline{j}$ is perpendicular to $\underline{y}$: $(a\underline{i} + b\underline{j}) \cdot (-\underline{i} + \sqrt{3} \underline{j}) = 0$ $-a + b\sqrt{3} = 0 \quad (1)$ The vector has a magnitude of 2: $\sqrt{a^2 + b^2} = 2$ $a^2 + b^2 = 4 \quad (2)$ Sub $a = b\sqrt{3}$ into (2): $(b\sqrt{3})^2 + b^2 = 4$ $3b^2 + b^2 = 4$ $4b^2 = 4$ $b^2 = 1$ $b = \pm 1$ When $b = 1 : a = \sqrt{3}$ When $b = -1 : a = -\sqrt{3}$ $\therefore$ The vector is $\sqrt{3}\underline{i} + \underline{j}$.	1

5. B	 $\underline{a} + \underline{b}$ and $\underline{a} - \underline{b}$ are the diagonals of a parallelogram. Since the diagonals are given as equal in length, the parallelogram must be a rectangle. Therefore $\underline{a}$ is perpendicular to $\underline{b}$.	1
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Section II	Suggested Solutions	Mk	Comments
6(a)	$\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{16-9x^2}} = \left[\frac{1}{3} \sin^{-1} \frac{3x}{4} \right]_0^{\frac{2}{3}}$ $= \frac{1}{3} \sin^{-1} \left(\frac{2}{4} \right) - 0$ $= \frac{1}{3} \times \frac{\pi}{6}$ $= \frac{\pi}{18}$	3	3Mk: Provides correct solution. 2Mk: Correctly finds primitive. 1Mk: Obtains $\sin^{-1} f(x)$.
6(b)	$7 \sin \theta - 2 \cos \theta = 5$ $7 \left(\frac{2t}{1+t^2} \right) - 2 \left(\frac{1-t^2}{1+t^2} \right) = 5$ $14t - 2 + 2t^2 = 5 + 5t^2$ $3t^2 - 14t + 7 = 0$ $t = \frac{14 \pm \sqrt{14^2 - 4(3)(7)}}{6}$ $\tan \frac{\theta}{2} = 4.097167541 \text{ or } 0.569499126$ $\frac{\theta}{2} = 76.28392^\circ \text{ or } 29.661475^\circ$ $\theta = 152^\circ 34' \text{ or } 59^\circ 19'$	3	3Mk: Provides correct solution. 2Mk: Correctly obtains one correct value of θ . 1Mk: Obtains correct quadratic equation in terms of t .
6(c)(i)	$\underline{a} = 5\underline{i} - 3\underline{j} \text{ and } \underline{b} = 3\underline{i} + 4\underline{j}$ $\underline{a} \cdot \underline{b} = (5)(3) + (-3)(4)$ $= 15 - 12$ $= 3$	1	1Mk: Provides correct solution.

6(c)(ii)	$\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{b} \cdot \vec{a}}{ \vec{a} ^2} \times \vec{a}$ $= \frac{3}{(\sqrt{5^2 + 3^2})^2} \times \vec{a}$ $= \frac{3}{34} (5\vec{i} - 3\vec{j})$ $= \frac{15}{34} \vec{i} - \frac{9}{34} \vec{j}$	2	2Mk: Provides correct solution. 1Mk: Obtains correct $ \vec{a} $.
6(d)(i)	$A \cos(x + \alpha) = A \cos x \cos \alpha - A \sin x \sin \alpha$ $A \cos \alpha = \sqrt{3}$ $A \sin \alpha = 1$ $A^2 = 3^2 + 1^2$ $A^2 = 4$ $A = 2$ $\tan \alpha = \frac{1}{\sqrt{3}}$ $\alpha = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$ $= \frac{\pi}{6}$ $\therefore \sqrt{3} \cos x - \sin x = 2 \cos \left(x + \frac{\pi}{6} \right)$	2	2Mk: Provides correct solution. 1Mk: Obtains A or α .
6(d)(ii)	$\sqrt{3} \cos x - \sin x = -1$ $2 \cos \left(x + \frac{\pi}{6} \right) = -1$ $\cos \left(x + \frac{\pi}{6} \right) = -\frac{1}{2}$ $x + \frac{\pi}{6} = \frac{-2\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3} \dots$ $x = -\frac{5\pi}{6}, \frac{\pi}{2}$	2	2Mk: Provides correct solution. 1Mk: Obtains one correct value of $x + \frac{\pi}{6}$.

7(a)	$x = y^{\frac{1}{3}}$ $V = \pi \int_0^4 x^2 dy$ $= \pi \int_0^4 y^{\frac{2}{3}} dy$ $= \pi \left[\frac{3y^{\frac{5}{3}}}{5} \right]_0^4$ $= \frac{3(4)^{\frac{5}{3}}}{5} \pi - 0$ $= 18.999...$ $= 19.0 \text{ u}^3 \text{ (Ignore rounding)}$	3	3Mk: Provides correct solution. 2Mk: Correctly integrates $y^{\frac{2}{3}}$ with a correct expression to find volume. 1Mk: Obtains the correct integrand to find volume (must contain π).
7(b)	$2 \cos^2 2x = 6 \sin^2 x - 1, \text{ for } 0 \leq x \leq 2\pi.$ $2 \cos^2 2x = 6 \sin^2 x - 1$ $2(1 - 2 \sin^2 x)^2 = 6 \sin^2 x - 1$ $2(1 - 4 \sin^2 x + 4 \sin^4 x) = 6 \sin^2 x - 1$ $2 - 8 \sin^2 x + 8 \sin^4 x - 6 \sin^2 x + 1 = 0$ $3 - 8 \sin^2 x + 8 \sin^4 x - 6 \sin^2 x = 0$ $8 \sin^4 x - 14 \sin^2 x + 3 = 0$ $8 \sin^4 x - 12 \sin^2 x - 2 \sin^2 x + 3 = 0$ $4 \sin^2 x (2 \sin^2 x - 3) - 1(2 \sin^2 x - 3) = 0$ $(2 \sin^2 x - 3)(4 \sin^2 x - 1) = 0$ $\sin^2 x = \frac{3}{2}$ $\sin x = \pm \sqrt{\frac{3}{2}}$ $x \text{ has no solutions}$ $4 \sin^2 x - 1 = 0$ $\sin^2 x = \frac{1}{4}$ $\sin x = \pm \sqrt{\frac{1}{4}}$ $\sin x = \pm \frac{1}{2}$ $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$	3	3Mk: Provides correct solution. 2Mk: Obtains a value for $\sin x$. 1Mk: Obtains a quadratic equation in $\sin^2 x$, $\cos^2 x$ or equivalent.

7(c)(i)	Resolving horizontal forces: $T_1 \cos 30 = T_2 \cos 45$ $T_1 = \frac{\left(\frac{1}{\sqrt{2}}\right)}{\left(\frac{\sqrt{3}}{2}\right)} T_2$ $= \frac{2}{\sqrt{6}} T_2$ $= \frac{\sqrt{6}}{3} T_2$	1	1Mk: Provides correct proof.
7(c)(ii)	Resolving vertical forces: $T_1 \sin 30 + T_2 \sin 45 = 12g$ $\frac{\sqrt{6}}{3} T_2 \sin 30 + T_2 \sin 45 = 12(10)$ $\frac{\sqrt{6}}{3} T_2 \times \frac{1}{2} + T_2 \times \frac{1}{\sqrt{2}} = 120$ $T_2 \left(\frac{\sqrt{6}}{6} + \frac{1}{\sqrt{2}} \right) = 120$ $T_2 = 120 \div \left(\frac{\sqrt{6}}{6} + \frac{1}{\sqrt{2}} \right)$ $= 107.589...$ $T_1 = \frac{\sqrt{6}}{3} \times 107.589...$ $= 87.846...$ $= 88\text{N (to nearest newton)}$	2	2Mk: Provides correct solution. 1Mk: Obtains a correct equation in terms of T_2 .

7(d)(i)	$\begin{aligned} \frac{d}{dx} x \tan^{-1}\left(\frac{x}{2}\right) &= x \left(\frac{\left(\frac{1}{2}\right)}{1 + \left(\frac{x}{2}\right)^2} \right) + \tan^{-1}\left(\frac{x}{2}\right) \\ &= \frac{x}{2} \div \left(1 + \frac{x^2}{4}\right) + \tan^{-1}\left(\frac{x}{2}\right) \\ &= \frac{x}{2} \div \left(\frac{4 + x^2}{4}\right) + \tan^{-1}\left(\frac{x}{2}\right) \\ &= \frac{x}{2} \times \left(\frac{4}{4 + x^2}\right) + \tan^{-1}\left(\frac{x}{2}\right) \\ &= \frac{4x}{8 + 2x^2} + \tan^{-1}\left(\frac{x}{2}\right) \\ &= \frac{2x}{4 + x^2} + \tan^{-1}\left(\frac{x}{2}\right) \end{aligned}$	2	2Mk: Provides correct solution. 1Mk: Correctly differentiates $\tan^{-1}\left(\frac{x}{2}\right)$.
7(d)(ii)	$\begin{aligned} \frac{d}{dx} x \tan^{-1}\left(\frac{x}{2}\right) &= \frac{2x}{4 + x^2} + \tan^{-1}\left(\frac{x}{2}\right) \\ \int \frac{d}{dx} x \tan^{-1}\left(\frac{x}{2}\right) dx &= \int \frac{2x}{4 + x^2} + \tan^{-1}\left(\frac{x}{2}\right) dx \\ \int \tan^{-1}\left(\frac{x}{2}\right) dx &= \int \frac{d}{dx} x \tan^{-1}\left(\frac{x}{2}\right) dx - \int \frac{2x}{4 + x^2} dx \\ &= x \tan^{-1}\left(\frac{x}{2}\right) - \ln(4 + x^2) + C \end{aligned}$	2	2Mk: Provides correct solution. 1Mk: Makes a valid attempt to use part (i) to write an integrand for $\tan^{-1}\left(\frac{x}{2}\right)$.

8(a)	$\int \cos 4x \cos x \, dx$ $= \int \frac{1}{2} [\cos(4x - x) \cos(4x + x)] \, dx$ $= \int \frac{1}{2} [\cos(3x) + \cos(5x)] \, dx$ $= \frac{\sin 3x}{6} + \frac{\sin 5x}{10} + C$	2	2Mk: Provides correct solution. 1Mk: Correctly applies the product to sum formula.
8(b)	$u = x^2 + 5$ $du = 2x \, dx$ and $x^2 = u - 5$ when $x = 2$, $u = 9$ and $x = 0$, $u = 5$ $\int_0^2 \frac{2x^3}{\sqrt{x^2 + 5}} \, dx = \int_5^9 \frac{2x(x^2)}{\sqrt{x^2 + 5}} \, dx$ $= \int_5^9 \frac{u - 5}{\sqrt{u}} \, du$ $= \int_5^9 u^{\frac{1}{2}} - 5u^{-\frac{1}{2}} \, du$ $= \left[\frac{2u^{\frac{3}{2}}}{3} - 10u^{\frac{1}{2}} \right]_5^9$ $= \left[\frac{2}{3} \left(9^{\frac{3}{2}} \right) - 10 \left(9^{\frac{1}{2}} \right) \right] - \left[\frac{2}{3} \left(5^{\frac{3}{2}} \right) - 10 \left(5^{\frac{1}{2}} \right) \right]$ $= [18 - 30] - \left[\frac{10\sqrt{5}}{3} - 10\sqrt{5} \right]$ $= \frac{20\sqrt{5}}{3} - 12 \text{ or } \frac{20\sqrt{5} - 36}{3}$	3	3Mk: Provides correct solution. 2Mk: Correctly integrates the expression with correct limits. 1Mk: Correctly obtains the integrand in u .
8(c)	Prove $2^n + 3^n - 5^n = 6M$, where $M \in \mathbb{Z}$ Prove true for $n = 1$: LHS = $2^1 + 3^1 - 5^1$ $= 0$ which is divisible by 6 The statement is true for $n = 1$. Assume true for $n = k$: where $k \in \mathbb{Z}^+$ i.e. $2^k + 3^k - 5^k = 6P$, where $P \in \mathbb{Z}$ i.e. $5^k = 2^k + 3^k - 6P$	3	3Mk: Provides correct solution. 2Mk: Successfully does 3 of 4 key parts below. 1Mk: Successfully does 2 of 4 key parts below.

	Prove true for $n = k + 1$: i.e. $2^{k+1} + 3^{k+1} - 5^{k+1} = 6Q$, where $Q \in \mathbb{Z}$ $ \begin{aligned} LHS &= 2^{k+1} + 3^{k+1} - 5^{k+1} \\ &= 2^{k+1} + 3^{k+1} - 5(2^k + 3^k - 6P) \\ &= 2(2^k) + 3(3^k) - 5(2^k) - 5(3^k) + 30P \\ &= -3(2^k) - 2(3^k) + 30P \\ &= -6(2^{k-1}) - 6(3^{k-1}) + 30P \\ &= 6(5P - 2^{k-1} - 3^{k-1}) \\ &= 6Q, \text{ where } Q = 5P - 2^{k-1} - 3^{k-1} \text{ is clearly an integer} \\ &\text{ since } k \text{ is a positive integer.} \end{aligned} $ The statement is true for $n = k + 1$ if true for $n = k$. Hence if it is true for $n = 1$, then it is true for $n = 2, 3$ and so on. $\therefore$ By the principle of mathematical induction, the statement is true for all positive integers n.		Key parts: 1. Proves true for $n = 1$. 2. Clearly states assumption and what must be proven. 3. Correctly uses assumption. 4. Correctly proves required statement.
8(d)(i)	$ \begin{aligned} \tan 3x &= \tan(2x + x) \\ &= \frac{\tan 2x + \tan x}{1 - \tan 2x \tan x} \\ &= \frac{\frac{2 \tan x}{1 - \tan^2 x} + \tan x}{1 - \frac{2 \tan x}{1 - \tan^2 x} \tan x} \\ &= \frac{\left(\frac{2t + t - t^3}{1 - t^2} \right)}{\left(\frac{1 - t^2 - 2t^2}{1 - t^2} \right)} \\ &= \frac{3t - t^3}{1 - 3t^2} \end{aligned} $	2	2 Mk: Provides correct proof. 1Mk: Uses compound angle formula to express $\tan 3x$ in terms of $\tan 2x$ and $\tan x$.
8(d)(ii)	$ \begin{aligned} \tan 3x \tan x &= 1 \\ \left(\frac{3t - t^3}{1 - 3t^2} \right) t &= 1 \\ 3t^2 - t^4 &= 1 - 3t^2 \\ t^4 - 6t^2 + 1 &= 0 \end{aligned} $	1	1Mk: Provides correct proof.

8(d)(iii)	$\tan 3x \tan x = \frac{\sin 3x \sin x}{\cos 3x \cos x}$ $= \frac{\frac{1}{2} [\cos(3x - x) - \cos(3x + x)]}{\frac{1}{2} [\cos(3x - x) + \cos(3x + x)]}$ $= \frac{\cos 2x - \cos 4x}{\cos 2x + \cos 4x}$ $\tan 3x \tan x = 1$ $\frac{\cos 2x - \cos 4x}{\cos 2x + \cos 4x} = 1$ $\cos 2x - \cos 4x = \cos 2x + \cos 4x$ $2 \cos 4x = 0$ $4x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \dots$ $x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8} \dots$ $\therefore$ The solutions to $x^4 - 6x^2 + 1 = 0$ are $\tan \frac{\pi}{8}, \tan \frac{3\pi}{8}, \tan \frac{5\pi}{8}, \tan \frac{7\pi}{8} \dots$ Also the solutions to $x^4 - 6x^2 + 1 = 0$ are $x^2 = \frac{6 \pm \sqrt{36 - 4(1)}}{2}$ $= \frac{6 \pm \sqrt{32}}{2}$ $= \frac{6 \pm 4\sqrt{2}}{2}$ $= 3 \pm 2\sqrt{2}$ $\therefore x = \pm \sqrt{3 \pm 2\sqrt{2}}$ Equating the smallest positive solution: $\tan \frac{\pi}{8} = \sqrt{3 - 2\sqrt{2}}$	3	3Mk: Provides correct solution. 2Mk: Correctly solves for $\tan 3x \tan x = 1$. 1Mk: Correctly applies a product to sum formula or obtains $x = \pm \sqrt{3 \pm 2\sqrt{2}}$.
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